Digital Mohalla — Layer diagram, iteration 1 (UML package diagram; design step A8, v1.2, as built 14 October 2026)

Layers with dependency inversion: each upper layer uses only the layer directly below it; the domain layer has no I/O and depends on no other layer; infrastructure implements the ports (store and repository interfaces) declared in the application layer, and uses the platform.

«layer» **Presentation**

- **Mobile app: resident and lender screens**
  W1–W2 request and decision, W3a–W4 handover and return, W5 dispute
- **Mobile app: moderator and admin screens**
  W6 case queue, W7 case view and decision; audit log view (screen to be designed)

One Expo / React Native app for every role, with screens chosen by role (NFR-D-05); Android and web.

«use»

«layer» **API / interface**

- **REST endpoints /api/v1**
  Architecture_Design.md section 4; JSON over HTTPS
- **Auth & authorisation**
  session, role, community scope (NFR-S-02, S-03)
- **Validation & error mapping**
  422 invalid input, 403 not allowed, 409 invalid transition / stale / dates taken
- **Privacy filter**
  no address, coordinates or CNIC in any response (NFR-PR-01)

«use»

«layer» **Application (use-case services)**

- **BorrowingService (M3)**
  B5–B8, B12 command handlers; ports LifecycleStore, AuditLog
- **CaseService, DecisionService (M9)**
  D6, I4
- **AuditLogService (M9)**
  I6
- **Auth & Membership (M1)**
  login, roles, coarse locality outcome
- **ScenarioSuite runner (M10)**
  CI only

«use»

«layer» **Domain (pure logic, no I/O)**

- **Lifecycle model**
  14 states, transition table T01–T26, guards; BorrowingTransaction + replay
- **Assurance engine (M5)**
  Beta posterior, decision policy, explanation, manipulation checks
- **Indirect trust (M5)**
  trust graph, bounded k-hop paths, noisy-OR, reliability cap
- **EscalationCase model**
  case states C01–C11

«realize» ports

«use»

«layer» **Infrastructure**

- **EventStore & projections**
  lifecycle_event, borrowing_transaction, replay_state()
- **Booking constraint**
  EXCLUDE on held bookings (ADR-0002)
- **Repositories**
  assessment, outcome, trust_edge, case, decision, audit_log
- **ConfigLoader**
  versioned engine config, config_version (NFR-D-01)
- **Notification adapter**
  planned, iteration 2 (not built)

«use»

«layer» **Platform (as built)**

- **PostgreSQL 16 + btree_gist**
  on Neon (AWS Singapore); schema.sql v2.3 via Alembic migration 0001
- **Render**
  API as a Docker web service, web app as a static site; deploys main after CI passes
- **GitHub Actions**
  ci.yml: lint, types, security scan, backend and frontend tests, API client check
- **Local run**
  PostgreSQL 16, Uvicorn, Expo dev server: development and demo fallback

«crosscutting» **Policies**

- **Authorisation on every transition**
  NFR-S-02; recorded in the event
- **Audit logging**
  append-only, hash-chained (FR-M-05)
- **Event schema**
  frozen fields + schema_version (FR-B-09)
- **Configuration**
  config_version on every assessment
- **Location privacy**
  coarse outcome only (NFR-PR-01/02)
- **Synthetic data labels**
  seed data marked synthetic
- **Logging**
  no request bodies with location; no secrets

UML package diagram: folder = package with stereotype «layer»; dashed open arrow = dependency («use» downward; «realize» from infrastructure up to the ports it implements). Border colour of an element = owner (purple S3, blue S1, orange S2, grey shared); grey dashed = planned, not built.